



Comprehensive Revision Notes: Modular Arithmetic and GCD

Introduction

This class covers the concepts of modular arithmetic and the greatest common divisor (GCD). These concepts are fundamental in computer science for optimizing and simplifying numerical computations.

Modular Arithmetic

Modular arithmetic involves using the modulo operation alongside arithmetic operations such as addition, subtraction, and multiplication. These functionalities help manage numerical ranges and avoid overflow in computations.

The Modulo Operator

- **Definition:** Given two numbers, a and b , $a \bmod b$ is the remainder after dividing a by b .
- **Range:** For $a \bmod b$, the result lies within $[0, b - 1]$ irrespective of a 's value. This is because the remainder can never exceed the divisor and can't be negative `【4:0+source】`.

Importance of the Modulo Operator

- Used to restrict values within a certain range, providing solutions to overflow issues.
- Important for ensuring computations like a^b remain within the acceptable range for data types `【4:4+source】`.

Modular Arithmetic with Basic Operations

1. **Addition:** $(A + B) \bmod M = ((A \bmod M) + (B \bmod M)) \bmod M$.



3. **Multiplication:** $(A \cdot B) \bmod M = ((A \bmod M) \cdot (B \bmod M)) \bmod M$.

4. **Exponentiation:** $(A^B) \bmod M = ((A \bmod M)^B) \bmod M$
 【4:6+source】 【4:11+source】 .

Use Cases

Besides avoiding overflow, modular arithmetic helps in cryptography, hash functions, and cyclic data structures.

Greatest Common Divisor (GCD)

The GCD of two numbers is the largest positive integer that divides both numbers without leaving a remainder.

Properties of GCD

1. **Commutative Property:** $GCD(A, B) = GCD(B, A)$.
2. **Associative Property:** $GCD(A, B, C) = GCD(GCD(A, B), C)$
 【4:18+source】 .
3. **GCD with Zero:** $GCD(A, 0) = |A|$. If one number is zero, the GCD is the absolute value of the other number 【4:18+source】 .

Euclidean Algorithm

The Euclidean algorithm efficiently finds the GCD of two numbers:

- $GCD(A, B) = GCD(B, A \bmod B)$.
- Continues until the second number becomes zero 【4:3+source】
 【4:12+source】 .

Use in Problem Solving

GCD is used in problems involving ratios, simplification of fractions, and solving linear diophantine equations among others.

Key Examples and Exercises



- $(A + B) \bmod M = 2$, using transformed form $((A \bmod M) + (B \bmod M)) \bmod M$.
- Multiply A and B , get results modulo M as $(A \cdot B) \bmod M = 2$
【4:4+source】 【4:11+source】 .

2. Using GCD for Simplification:

- Example: Find GCD of 15 and 6:
 - Sequence: $GCD(15, 6) = GCD(6, 3) = GCD(3, 0) = 3$
【4:12+source】 【4:14+source】 .

Conclusion

Modular arithmetic and GCD are essential for efficient computations and problem-solving in software engineering. Understanding their principles and applications can significantly improve algorithm implementation and optimization.

Further Reading

For those interested in deeper exploration, consider algorithms involving modular multiplicative inverses and extended Euclidean algorithms. These are especially important in fields like cryptography and computational number theory.